Temporal Analysis and Assessment of Reference Evapotranspiration (ET_0) in Kutch, Gujarat Using NASA POWER Data with Google Earth Engine Based Validation

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Abstract— Reference evapotranspiration ET_0 is a significant parameter for irrigation scheduling, estimation of crop water requirement and sustainable water-resource management in arid and semi-arid regions. The Kutch District of Gujarat is characterized by high climatic variability, low and erratic rainfall, high temperatures and low availability of water. The present study investigates the temporal variability of reference evapotranspiration (ET_0) for Kutch using hourly meteorological data from the NASA Prediction of Worldwide Energy Resources (NASA POWER) database for the period of January 2010 to February 2022. ET_0 was estimated by the FAO-56 Penman–Monteith approach using meteorological variables including air temperature, relative humidity, wind speed and radiation related parameters. The hourly ET_0 values from the model were aggregated to daily values and analyzed at hourly, monthly and seasonal scales. Furthermore, the daily ET_0 estimates from the NASA POWER were compared to the correspondent daily ET_0 values from a Google Earth Engine (GEE)-derived dataset for the common period from 1 January 2010 to 20 February 2022. The comparison was based on the coefficient of determination (R2), root mean square error (RMSE), mean absolute error (MAE), and mean bias error (MBE). The validation showed R2 of 0.6312, RMSE of 1.2692 mm/day, MAE of 1.0238 mm/day and MBE of -0.1607 mm/day. The NASA POWER analysis revealed that the mean daily ET_0 was higher during pre-monsoon period, with April and May recording about 9.45 and 10.19 mm/day respectively. Whereas, the seasonal mean during summer/pre-monsoon period was about 9.25 mm/day. The results reveal the temporal variability of the atmospheric evaporative demand and can be useful for irrigation planning and water-resource management in the arid region of Kutch.

Keywords— Reference Evapotranspiration, ET_0, FAO-56 Penman–Monteith, NASA POWER, Google Earth Engine, Kutch, Gujarat, Temporal Variability, Irrigation Planning, Water Resource Management

I. Introduction

Evapotranspiration (ET_0) Total water loss to the atmosphere from evaporation of soil and water surfaces and transpiration from vegetation. It is an important component of the hydrological cycle and has a major impact on agricultural water management, irrigation scheduling, estimation of crop water requirements and water resource planning. Accurate estimation of evapotranspiration is of particular importance in areas where water availability is limited and agricultural activities are highly influenced by climatic conditions.

Reference evapotranspiration (ET_0) is the rate of evapotranspiration of a hypothetical reference surface under specified meteorological conditions. It offers a standardized measure of atmospheric evaporative demand and can be used as the basis for estimating crop water requirements when paired with appropriate crop coefficients. One of the methods for estimating ET_0 is the FAO-56 Penman-Monteith method, which is widely used because it incorporates the effects of temperature, humidity, wind speed, and radiation into a physically based formulation

The temporal analysis of ET_0 is helpful for understanding the variation of atmospheric water demand in different time scales. Hourly observations can give information on diurnal variations, and daily, monthly and seasonal analyzes can show broader patterns associated with climatic conditions. Such information is useful for better irrigation management and for efficient use of available water resource

A. Area of Study and Problem

The climate of Kutch District in western Gujarat, India is generally arid to semi-arid with high temperatures, variable rainfall and considerable spatial and temporal variability in climatic conditions. Strong seasonal variations in atmospheric conditions are observed in the region, which can influence the evaporative demand of the atmosphere and, consequently, the agricultural water requirement.

Water availability is a critical issue in arid and semi-arid environments. Temperature, humidity, wind speed and radiation variations may result in substantial changes of reference evapotranspiration during a day and a season. So, quantifying temporal behaviour of ET_0 in Kutch may provide useful information for understanding the atmospheric water demand and for supporting irrigation and water-resource management. The location of the study area is shown in fig.1.



Fig. 1. Location of the study area in Gujrat.

The present study is concerned with the temporal assessment of reference evapotranspiration in Kutch using meteorological data obtained from the NASA Prediction of Worldwide Energy Resources (NASA POWER) database. The analysis considers the available period from January 2010 to February 2022 and examines ET_0 at different temporal scales

B.Utilization of Meteorological and Gridded Data

Reliable meteorological observations are required for reference evapotranspiration estimation. But long-term station-based observations may not be readily available with sufficient spatial and temporal resolution. Gridded and satellite-supported meteorological datasets can also be used to inform climate and environmental studies.

NASA POWER is a provider of meteorological and solar energy data for environmental and agricultural applications. In this study meteorological variables such as air temperature, relative humidity, wind speed and radiation related parameters were used for estimation and analysis of ET_0. The data were processed at hourly intervals and then aggregated to daily, monthly and seasonal levels.

Apart from the NASA POWER based analysis, a corresponding ET_0 dataset derived from Google Earth Engine (GEE) was used for comparison. The GEE derived data provided an independent dataset to evaluate the consistency of the NASA POWER based daily ET_0 estimates during their common period of availability. The comparison was conducted using statistical validation metrics like coefficient of determination (R^2), root mean square error (RMSE), mean absolute error (MAE) and mean bias error (MBE).

C. Research Gap and Objectives.

While reference evapotranspiration has been studied using different meteorological data sets and estimation techniques, detailed temporal assessment at multiple time scales is still important for arid regions like Kutch. Hourly, monthly and seasonal variations in atmospheric conditions can result in substantial changes in evaporative demand. Comparison with an independently derived dataset can also provide further information on the consistency of ET_0 estimates.

II. LITERATURE SURVEY

A. FAO-56 Penman-Monteith Method

The FAO-56 Penman–Monteith approach is a popular way to calculate reference evapotranspiration (ET_0). The method takes into account the combined effects of air temperature, relative humidity, wind speed and available energy on evapotranspiration. Its standardized formulation enables the inter-comparison of ET_0 between different climatic conditions and locations. The FAO-56 Penman-Monteith method has been used as a reference method in previous studies for agricultural water management and irrigation planning.

B. Meteorological Data and Gridded Data for ET_0 Estimation

Long-term meteorological data are important for reliable estimation of ET_0. Datasets that are satellite-supported and gridded can be a useful alternative to conventional weather-station observations in data-poor regions. NASA POWER has been used for a variety of agriculture and environment applications because of the meteorological and solar-related variables available at suitable temporal scales. Such datasets allow for ET_0 estimation and temporal analysis in areas where continuous ground observations may not be available.

Google Earth Engine (GEE) has also proven useful in processing and analyzing large environmental datasets. The cloud processing capabilities allow the access and analysis of meteorological and satellite derived information over specific geographical regions. Therefore, the consistency of the results can be examined comparing ET_0 estimates from different data sources.

C. ET_0 Variability in Arid and Semi-Arid Regions

In studies of ET_0 in arid and semi-arid environments seasonal variations in temperature, humidity, wind speed and radiation have been found to be strong determinants of ET_0. Warmer and drier periods have high atmospheric demand and potentially higher reference evapotranspiration, while relatively cooler and more humid conditions tend to lead to lower values. These temporal variations are of particular importance for irrigation scheduling and efficient water-resource management in water-limited regions.

The analysis of multi-temporal ET_0 can be useful for knowing the changes in the atmospheric water demand in a region like Kutch where rainfall is variable and water availability is limited. Hourly analysis can also detect diurnal patterns that may not be apparent in monthly or seasonal averages

D. Research Gap

Previous studies have demonstrated the usefulness of the FAO-56 Penman–Monteith method and gridded meteorological datasets for ET_0 estimation. However, the short-term multi-temporal assessment of ET_0 for Kutch region using NASA POWER data with independent comparison using Google Earth Engine derived dataset is required.

Thus, this study combines the hourly-to-daily processing, monthly and seasonal analysis and statistical validation of NASA POWER-based ET_0 estimates with the corresponding GEE-derived values. The validation provides quantitative measures of the agreement by R2, RMSE, MAE and MBE, thus extending the analysis from a descriptive temporal perspective.

The relevant studies and approaches considered in this work are summarized in Table I.

TABLE I. SUMMARY OF RELEVANT LITERATURE

Study / Approach	Data / Method	Major Focus
FAO-56 Penman–Monteith	Meteorological variables	Standardized ET_0 estimation
NASA POWER-based studies	Gridded meteorological data	ET_0 estimation and agricultural applications
Google Earth Engine-based studies	Cloud-based environmental datasets	Large-scale environmental and temporal analysis
Arid and semi-arid ET_0 studies	Temperature, humidity, wind and radiation	Seasonal and climatic variability of ET_0
Present Study	NASA POWER + GEE	Multi-temporal ET_0 analysis and statistical validation in Kutch

III. METHODOLOGY

The methodology used in this study includes data acquisition, pre-processing, reference evapotranspiration estimation, temporal aggregation and validation. NASA POWER meteorological data were the main data set, and the corresponding dataset derived from Google Earth Engine (GEE) was used for comparison and validation.

A. Study area

The study was conducted in Kutch District, Gujarat, India, an arid to semi-arid region with high temperatures, variable rainfall and limited water availability. Climate in the region results in large temporal variations in atmospheric evaporative demand. Hence, Kutch is a suitable region to study temporal behavior of reference evapotranspiration

B. NASA POWER Data

The hourly meteorological data of Kutch were retrieved from NASA Prediction of Worldwide Energy Resources (NASA POWER) database for the available period January 2010 to February 2022. The variables used in the analysis were air temperature, relative humidity, wind speed and radiation related parameters required for the estimation of ET_0.

Downloaded data were sorted according to date and time and then processed to obtain variables required for FAO-56 Penman–Monteith formulation. The NASA POWER dataset contains 106,423 hourly observations

C. Google Earth Engine Dataset

An independent ET_0 dataset obtained through Google Earth Engine was used for comparison with the NASA POWER-based estimates. The GEE dataset included hourly records and corresponding daily ET_0 values. The common period was used for validation only due to the different temporal coverage of the two datasets.

The standard validation period was from 1 January 2010 to 20 February 2022. After matching the dates and keeping complete daily observations, 4,434 daily observations were available for statistical comparison.

D. Data Preprocessing and ET_0 Estimation

Meteorological observations were checked and arranged in sequence before analysis. Hourly ET_0 values were computed using the FAO-56 Penman–Monteith method. The formulation used in this study is given as:

$$ET_0=\Delta+\gamma(1+0.34u2)0.408\Delta(Rn-G)+\gamma T+273900u2(es-ea)$$

Where:

ET_0 = Reference evapotranspiration (mm/day)

Δ= Slope of the saturation vapour pressure curve (kPa/°C)

R_n= Net radiation at the crop surface (MJ/m²/day)

G= Soil heat flux density (MJ/m²/day)

γ= Psychrometric constant (kPa/°C)

T= Mean air temperature at 2 m height (°C)

u_2= Wind speed at 2 m height (m/s)

e_s= Saturation vapour pressure (kPa)

e_a= Actual vapour pressure (kPa)

$e_s - e_a$= Vapour pressure deficit (kPa)

The daily values were then applied to calculate monthly and seasonal mean ET_0 . Seasonal analysis was performed for the winter, summer/pre-monsoon, monsoon and post-monsoon seasons.

E. NASA POWER–GEE Validation

To verify, daily ET_0 values from NASA POWER and the GEE-derived dataset were compared by date over the period of overlap. The daily ET_0 value of the GEE dataset was linked to its hourly records, thus the comparison was conducted using the daily value directly instead of counting the repeated value for several times.

The agreement between the two datasets was evaluated using the coefficient of determination ((R^2)), root mean square error (RMSE), mean absolute error (MAE), and mean bias error (MBE):

RMSE (Root Mean Square Error)

$$\mathbf{RMSE} = \sqrt{[(1/n)\ \Sigma_{i=1}^{n}\ (ET_{0,NASA,i} - ET_{0,GEE,i})^2]}$$

MAE (Mean Absolute Error)

$$MAE = (1/n)\ \Sigma_{i=1}^{n}\ |ET_{0,NASA,i} - ET_{0,GEE,i}|$$

MBE (Mean Bias Error)

$$MBE = (1/n)\ \Sigma_{i=1}^{n}\ (ET_{0,NASA,i} - ET_{0,GEE,i})$$

Where:

n = total number of matched observations

$ET_{0,NASA,i}$ = NASA POWER ET_0 for observation i

$ET_{0,GEE,i}$ = GEE ET_0 for observation i

RMSE, MAE, and MBE are expressed in mm/day.

IV. RESULTS AND DISCUSSION

A. Overall ET_0 Characteristics

The variation of reference evapotranspiration ET_0 with time is observed at various periods of the year in the Kutch region. The variation of ET_0 depends on the variations of air temperature, solar radiation, relative humidity and wind speed. The warmer and drier pre-monsoon period is generally associated with higher values of atmospheric evaporative demand, while the cooler and more humid periods with lower values.

The analysis of monthly ET_0 reveals that during the pre-monsoon period ET_0 increases and attains its maximum values during the months of April and May. The mean daily ET_0 for April and May from the NASA POWER dataset is approximately 9.45 mm/day and 10.19 mm/day, respectively. In contrast, lower monthly values are observed during winter and relatively humid periods. Higher ET_0 during pre-monsoon period may be related to higher temperature, availability of radiation and atmospheric dryness.

The monthly and seasonal variations of mean daily ET_0 are shown in Fig. 2.

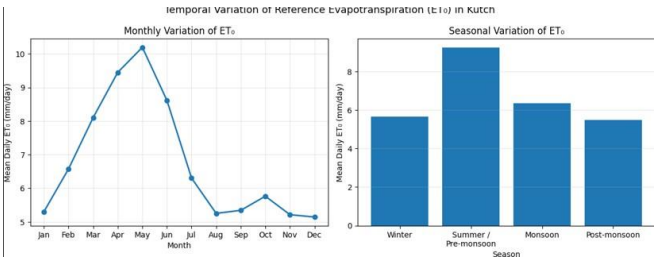


Fig. 2. Fig. 2. Monthly and Seasonal Variation of Mean Daily Reference Evapotranspiration (ET_0) in Kutch.

The maximum mean daily ET_0 is observed during the summer/pre-monsoon season and is about 9.25 mm/day. The mean values during winter, monsoon and post monsoon are approximately 5.66, 6.36 and 5.49 mm/day respectively. The results indicate a large seasonal variation in atmospheric evaporative demand in Kutch.

B. *Diurnal Variation*

Hourly analysis reveals a distinct diurnal pattern of reference evapotranspiration (ET_0) over Kutch. Figure 3 shows that ET_0 is relatively low during the night and early morning hours with values less than about 0.10 mm/hour. ET_0 increases rapidly from about 07:00 and reaches its maximum value of about 0.75 mm/hour at about 12:00. ET_0 decreases gradually during the afternoon and evening hours after midday.

The observed increase during the day is consistent with the increased availability of solar energy and changes in atmospheric conditions during the day time hours The reduction after midday reflects a decrease in the combined atmospheric demand as expressed in the ET_0 calculation. FAO defines ET_0 as a climatic evaporative-demand measure based on variables such as radiation, temperature, humidity, and wind speed.

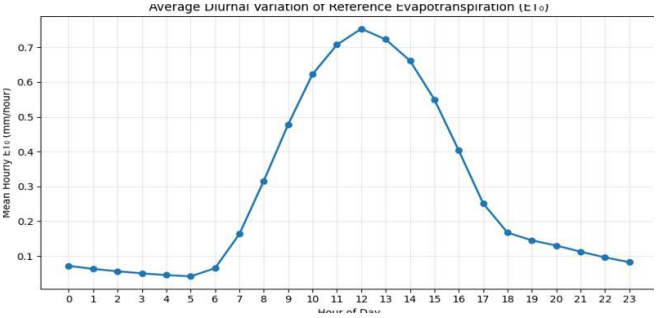


Fig. 3. Average Diurnal Variation of Reference Evapotranspiration (ET_0) in Kutch

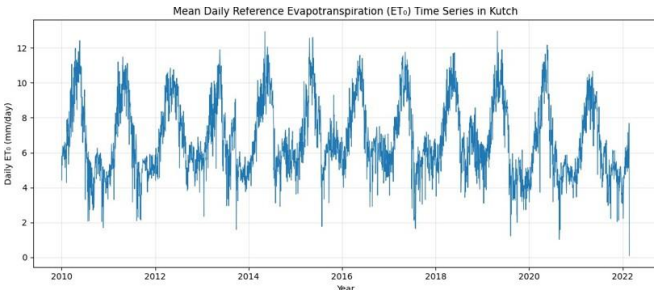


Fig. 4. Daily Reference Evapotranspiration (ET_0) Time Series in Kutch, 2010–2022

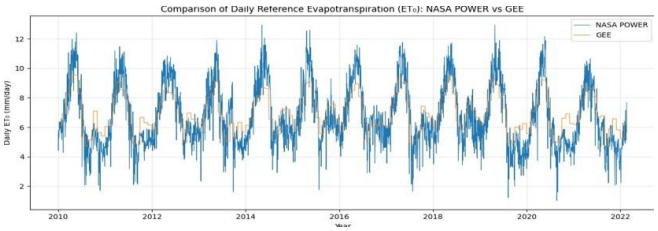


Fig. 5. Comparison of Daily Reference Evapotranspiration (ET_0) from NASA POWER and Google Earth Engine over the Common Validation Period (2010–2022).

C. *Validation of NASA POWER and Google Earth Engine ET_0 Estimates*

A validation analysis was carried out to evaluate the reliability of the NASA POWER based reference evapotranspiration (ET_0) estimates using ET_0 values obtained from Google Earth Engine (GEE). The comparison involved a total of 4,434 matched daily observations for the common period from January 2010 to February 2022.

Coefficient of determination (R2), root mean square error (RMSE), mean absolute error (MAE), and mean bias error (MBE) were used to evaluate the agreement between the two datasets. The statistical results indicate a fairly good agreement between the ET_0 estimates from the NASA POWER and GEE. The coefficient of determination (R^2 = 0.6312) indicates a moderate to strong positive relationship between the two datasets. The RMSE value of 1.2692 mm/day and MAE value of 1.0238 mm/day shows that the average difference between the two methods is relatively small. The negative value of MBE (−0.1607 mm/day) shows that the NASA POWER based ET_0 estimates slightly underestimate ET_0 values than the GEE based estimates.

The scatter plot analysis also suggests that the majority of the observations cluster around the 1:1 reference line, suggesting consistency between the two datasets. Small differences can be attributed to different meteorological inputs, spatial and temporal resolutions, and processing procedures used in the NASA POWER and GEE platforms. The validation results generally show that the NASA POWER meteorological data along with the FAO-56 Penman–Monteith method can give reliable estimates of reference evapotranspiration in the arid climatic conditions of the Kutch region.

TABLE II. STATISTICAL VALIDATION BETWEEN NASA POWER AND GEE ET_0 ESTIMATES

Validation Metric	Value
Number of Observations	4,434
Validation Period	2010–2022
R^2	0.6312
RMSE (mm/day)	1.2692
MAE (mm/day)	1.0238
MBE (mm/day)	-0.1607

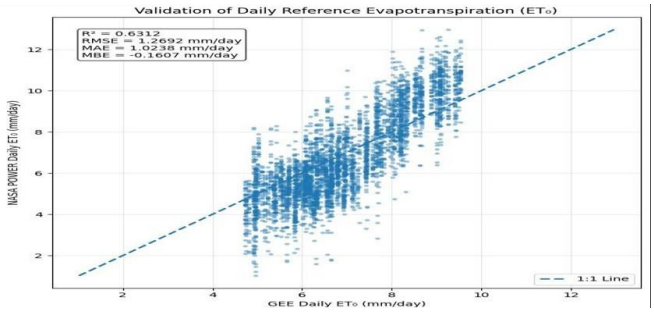


Fig. 6. Scatter plot showing the relationship between NASA POWER-based and Google Earth Engine-based daily reference evapotranspiration (ET_0) values during the validation period (2010–2022). The dashed line represents the 1:1 reference line.

D. Discussion

The results show a distinct temporal variation of reference evapotranspiration ET_0 over the Kutch region. The monthly analysis reveals that ET_0 increases from the winter months to the pre-monsoon period and attains its maximum mean values in April and May. This pattern indicates higher atmospheric evaporative demand during the warmer part of the year. The seasonal analysis also reveals the maximum mean daily ET_0 in summer/pre-monsoon season and minimum value in winter and post-monsoon periods. Furthermore, the diurnal analysis reveals that ET_0 mainly occurs in the daytime. After the early morning, the hourly ET_0 increases, reaching its maximum around noon, and then gradually decreasing during the afternoon and evening. This temporal pattern is in line with the variation of meteorological conditions that influence evapotranspiration, particularly radiation and atmospheric demand. The long-term daily time series presents a recurrent seasonal pattern over the whole observation period of NASA POWER. Higher ET_0 values are generally found during warmer months, whereas lower values are seen during cooler and more humid periods. However, there is also a large day-to-day variability, indicating that short-term meteorological conditions can cause large variations in atmospheric evaporative demand. Comparison between NASA POWER and GEE reveals that both datasets reproduce a broadly similar temporal pattern during the common validation period. Validation gave R2 of 0.6312 with RMSE and MAE of 1.2692 mm/day and 1.0238 mm/day respectively. The MBE of −0.1606 mm/day means that the NASA POWER estimates are biased slightly negative relative to the GEE values. The scatter distribution around the 1:1 line also indicates that the two datasets capture similar variations of daily ET_0, but differences are found for individual observations. These results emphasize the need for reliable meteorological data to evaluate the water demand of the atmosphere in arid regions. The temporal variability observed in ET_0 can be useful for assessment of irrigation requirement and water-resource planning in Kutch. However, the validation results also suggest that the ET_0 estimates should be understood in relation to the characteristics and processing techniques of the underlying datasets

V. CONCLUSION.

The present study examined the temporal variability of reference evapotranspiration (ET_0) for the Kutch region of Gujarat using the meteorological data from NASA POWER and the FAO-56 Penman-Monteith approach. The results indicated distinct monthly, seasonal and diurnal changes in ET_0 . Highest atmospheric evaporative demand was found in the summer/pre-monsoon period and lower values in the cooler periods. The NASA POWER estimates were also compared with the GEE-derived ET_0 values during the common period (January 2010–February 2022). The validation gave a R^2 of 0.6312, RMSE of 1.2692 mm/day, MAE of 1.0238 mm/day, and MBE of −0.1607 mm/day, which indicates a reasonable agreement between the two datasets with a slight negative bias in the NASA POWER estimates.

The results provide useful information for understanding atmospheric water demand of Kutch and can aid irrigation planning and water-resource management of this arid region.

The highest monthly mean ET_0 was observed in May (10.19 mm/day), while the summer/pre-monsoon season recorded the highest seasonal mean (9.25 mm/day).

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